ContextWorld: Benchmarking In-Context Dynamics Adaptation in Latent Visual World Models

Abstract

Latent world models predict future states from visual observations and action sequences within a learned representation space. When environment dynamics vary, accurate prediction requires inferring hidden dynamic properties from interaction history. However, because models can exploit immediate visual shortcuts or regress to average dynamics to achieve low loss without history conditioning, conventional prediction accuracy fails to reveal genuine adaptation. We introduce ContextWorld, a benchmark designed to evaluate in-context dynamics adaptation in action-conditioned latent world models under strictly frozen weights. ContextWorld employs paired counterfactual evaluations—varying interaction histories while holding the current observation and future action sequence fixed—to concurrently quantify adaptation accuracy (whether predictions correctly distinguish between diverging future states) and history sensitivity (whether predictions respond to altered context and yield lower error than mismatched history). Spanning nine tasks across four environments, the benchmark provides separate training and evaluation splits with continuous interaction rollouts under differing dynamics. Our systematic evaluation of representative latent world models (LeWM, PLDM, and DINO-WM) yields three key findings: First, models trained under fixed dynamics make negligible use of interaction history across most tasks. Second, mixing multi-dynamics data improves history responsiveness, but the benefits of data scaling and the relative effectiveness of training strategies vary substantially across tasks. Third, under certain training schemes, models fail to effectively leverage history even on the training distribution, despite steadily decreasing training loss. These results reveal a notable mismatch between prediction-loss optimization and in-context dynamics adaptation, suggesting that standard prediction objectives do not reliably drive models to use interaction history and highlighting the need for objectives and benchmarks explicitly targeted at in-context dynamics adaptation.

1. Introduction

Visual world models learn environment dynamics to predict future states [19], with latent world models performing these predictions within learned representations of visual observations [20, 21]. A model trained under fixed dynamics can achieve high prediction accuracy without utilizing interaction history to adapt its predictions to current conditions. However, such predictions can fail when environment dynamics vary [33]. For example, visually identical objects can move different distances under the same push if surface friction differs. Interaction history provides evidence about these hidden physical properties. We investigate whether latent world models can leverage this contextual evidence to predict future states under varied dynamics *without updating their weights*—an ability we term *in-context dynamics adaptation*.

Physical prediction benchmarks evaluate whether observed interactions help models infer hidden properties or predict downstream outcomes governed by them [4, 32]. While benchmarks like Physion++ assess physical understanding via auxiliary readout classifiers, we introduce ContextWorld (Figure 1) to evaluate in-context dynamics adaptation directly within action-conditioned latent world models, leveraging their existing latent predictors without training separate classifiers for future events. The benchmark spans nine tasks across four environments, covering physical parameter variations and interaction rules, and provides separate training and evaluation data.

In its paired tasks, two dynamics conditions share the identical current observation and future action sequence, but produce diverging ground-truth futures. Interaction history is therefore strictly necessary to distinguish the two outcomes. Because a deterministic model that ignores history receives identical inputs and yields identical predictions for both conditions, it scores at most 50% on paired future selection when ties count as failures. This paired counterfactual design restricts history-agnostic heuristics, enabling us to concurrently evaluate two aspects directly in latent space: (1) adaptation accuracy, which measures whether a model's prediction correctly aligns with its corresponding true future rather than the alternative condition's future; and (2) history sensitivity, which measures whether predictions re-

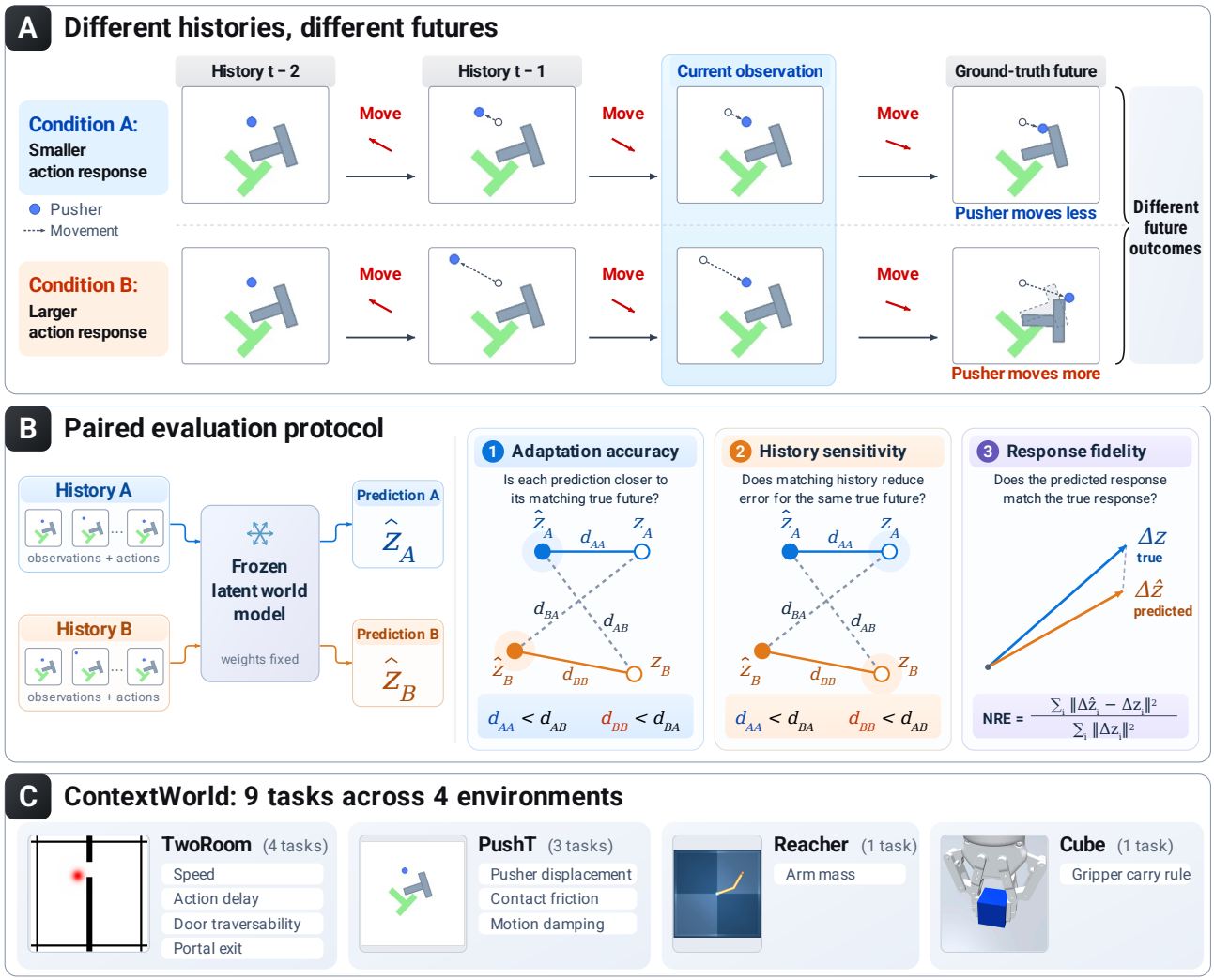


Figure 1. **ContextWorld overview.** (A) A recorded pusher-displacement pair with shared current observations and future actions. Red arrows show action directions; dashed arrows mark observed endpoint displacement. The translucent outline in the lower future panel marks the T block's current pose. These annotations are not model inputs; all scene images use the same crop of the original observations. (B) The frozen model produces latent predictions from each history. The three subpanels illustrate complementary checks for paired tasks. Filled circles denote predictions and open circles denote encoded true futures. The first two views share the same four points. Halos mark predictions in adaptation accuracy and true futures in history sensitivity, indicating the fixed reference for each comparison. Here d_{ij} is the error between the prediction from history i and true future j; the inequalities show the desired ordering for both conditions. Response fidelity compares $\Delta\hat{z} = \hat{z}_B - \hat{z}_A$ with $\Delta z = z_B - z_A$, using the same schematic points. All latent-space geometry is illustrative, not a projection of measured model outputs. (C) Nine tasks across four environments.

liably respond to altered context, verified by substituting the history with that from an alternate dynamics setting to examine if matching history yields strictly lower error.

Using ContextWorld, we benchmark three representative latent world models: LeWM [24], PLDM [30], and DINO-WM [37]. We find that after training under fixed dynamics, these models make negligible use of interaction history across most tasks. To examine how this adaptation ability can be acquired, we mix ContextWorld training data with original rollouts and compare training from scratch against fine-tuning models pretrained under fixed dynamics. We further scale up the training data and examine the impact of freezing the visual encoder. Our experiments reveal that while data scaling improves history utilization on several tasks, the optimal model and training strategy vary substantially by task. For example, starting from the same pretrained weights, freezing PLDM's visual encoder improves its ability to distinguish future states under varying contact

friction, but reduces this ability under motion damping.

To understand why models often fail to leverage history, we analyze model predictions on the training distribution itself. Under certain training schemes, training loss steadily decreases or approaches convergence even while models fail to use interaction history to distinguish physical conditions. Because this failure occurs on the training data, it cannot be explained by out-of-distribution generalization alone, suggesting that standard prediction objectives may not inherently provide sufficient incentive for models to condition their predictions on interaction history.

Our main contributions are:

- **The ContextWorld benchmark and paired evaluation protocol.** We introduce a standardized framework featuring separate training and evaluation data across nine hidden-dynamics tasks. This paired design quantifies both adaptation accuracy and history sensitivity directly from action-conditioned latent rollouts.
- **Empirical findings on adaptation limits.** We benchmark representative models across data scales, training strategies, and visual encoder freezing schemes, demonstrating that optimal strategies are highly task-dependent. Furthermore, we show that training loss can steadily decrease even when models fail to leverage history on the training data, revealing a disconnect between standard prediction losses and in-context dynamics adaptation.

2. Related Work

Visual world models. Visual world models learn environment dynamics to simulate and predict future states. Early frameworks [19] combined learned visual representations with recurrent dynamics to support internal policy training. Subsequent methods advanced this paradigm through online planning (PlaNet [20]) and policy learning via imagined latent trajectories (Dreamer [21]). While these early approaches rely on pixel-level image reconstruction to learn representations, Joint-Embedding Predictive Architectures (JEPAs) [23] shift the training objective toward predicting target representations directly without reconstructing pixels. This principle has been applied to masked spatial regions (I-JEPA [2]) and extended to temporal video representation learning (V-JEPA [5]). To support robotic planning, V-JEPA 2 [3] introduces an action-conditioned predictor trained on robot interactions atop pretrained video representations.

Action-conditioned latent world models also differ in how they obtain their visual representations. DINO-WM [37] trains a predictive dynamics model on frozen pretrained features for planning from offline trajectories. In contrast, PLDM [30] jointly optimizes the visual encoder and the predictor using reward-free offline data. To stabilize joint training, LeWM [24] simplifies the objective to a latent prediction loss and a single representation regularizer. We benchmark these three paradigms to evaluate how structural choices influence in-context dynamics adaptation.

In-context adaptation. When environment dynamics differ between training and deployment, interaction history provides a signal for adaptation without requiring weight updates. Meta-reinforcement learning [6] leveraged this principle by training agents to adapt from limited experience across tasks. RL^2 [14] implicitly stores interaction history within a recurrent network's hidden state, while PEARL [29] infers a task representation from experience to condition the policy. Paralleling large language models, which learn from in-context examples without parameter updates [8, 13], this adaptation paradigm has been extended to visual world models.

Within world models, context guides predictions of action outcomes in new environments. OSWM [16] pretrains a transformer on synthetic dynamics, using observed state transitions as context to forecast futures. L2World [33] distinguishes using context to identify previously seen environments from learning new dynamics. It studies the effects of context length and training environment diversity through multi-step prediction errors and context perturbations. For robotic control, ICWM [34] employs task-agnostic interaction history to adapt policies to novel camera viewpoints without weight updates, using ablations over correct, absent, and mismatched contexts to validate history reliance.

Benchmarks. To evaluate adaptation across environments, existing benchmarks introduce controlled variations in dynamics, appearance, or task rules. Meta-World [35] and Procgen [11] test generalization across manipulation tasks and procedurally generated levels. CausalWorld [1] and CARL [7] vary environment properties, while the XLand series (XLand-MiniGrid [26], XLand-100B [27]) evaluates policy adaptation from interaction history across distributions of tasks.

For physical prediction, several benchmarks focus on counterfactual reasoning and hidden property inference. CoPhy [4] asks models to predict outcomes after changes to a scene's initial configuration, using the original trajectory as evidence of hidden properties. Filtered-CoPhy [22] ensures the observed trajectory contains sufficient information for the counterfactual prediction. ComPhy [9] probes reasoning regarding hidden mass and charge through predictive and counterfactual video querying. Physion++ [32] tests whether interaction history helps predict future contact in paired trials that share the same current observation but differ in physical properties, assessing this understanding via auxiliary readout classifiers.

Beyond visual domains, a recent time-series benchmark [36] reveals that low prediction error can coexist with incorrect response directions when actions change. This

echoes our finding of a mismatch between prediction-loss minimization and true dynamics adaptation. A remaining question is when action-conditioned latent world models learn to leverage history for accurate prediction. ContextWorld addresses this gap through controlled comparisons across models, tasks, data scales, and training strategies.

3. ContextWorld Benchmark

ContextWorld is designed to evaluate in-context dynamics adaptation—specifically, whether models use interaction history to accurately predict future states conditioned on an action sequence. This section formalizes the prediction setting, describes how we construct the tasks and datasets, and explains the paired evaluation protocol and metrics.

3.1. Problem Formulation

To isolate in-context adaptation from weight updates, model parameters θ remain fixed during evaluation. The model receives exclusively visual observations and actions. For a target horizon τ, the model receives an interaction history H_t, comprising interleaved visual observations up to time t and their executed actions, alongside a future action sequence $A_{t:t+\tau-1}$. The prediction and its target are

$$\begin{aligned} \hat{z}_{t+\tau} &= f_\theta(H_t, A_{t:t+\tau-1}), \\ z_{t+\tau} &= e_\theta(o_{t+\tau}). \end{aligned} \tag{1}$$

Here, $o_{t+\tau}$ is the ground-truth observation at time $t+\tau$, and e_θ is the model's target encoder, which is also frozen. Predictions are evaluated directly against these encoded targets in latent space.

3.2. Task Design

To evaluate dynamics adaptation, we construct nine tasks spanning four base environments by introducing controlled variations in physical parameters and interaction rules. These environments build upon TwoRoom navigation [37], the PushT pushing environment [10], Reacher from the DeepMind Control Suite [31], and Cube from OGBench [28]. Figure 1 illustrates the task structure and paired comparisons; Table 1 summarizes the tasks and their history requirements. Importantly, models are trained separately for each task, and baselines like DINO-WM keep their pretrained visual encoder frozen throughout.

To isolate the role of interaction history, we construct paired counterfactual samples for six tasks: pusher displacement, arm mass, contact friction, motion damping, gripper carry rule, and portal exit. Within each pair, the two samples share the same current observation and future action sequence, but operate under different hidden dynamics. The history and ground-truth future for each sample are generated through one continuous simulation with fixed dynamics. Because the resulting futures differ physically, accurate prediction necessitates conditioning on the interaction history. Trajectory pairing and validity checks are detailed in Appendix B.

Unlike the paired counterfactuals, the remaining three tasks evaluate history usage across multiple condition values or specific environmental rules. The speed task tests generalization to unseen velocities: we hold the current observation and planned actions fixed, providing histories generated at varying speeds to observe if the model scales its predicted motion. Action delay introduces variable latency to test predictions before and after an action takes effect. For example, delays of six or ten simulator steps both result in zero movement within the next five steps. Because these yield identical future states, they are scored as a single response group, with all groups weighted equally (Appendix C). Finally, the door traversability task tests whether the model correctly infers passage or blockage from preceding interactions.

3.3. Dataset Construction and Splits

To evaluate models on novel interaction rollouts, all tasks provide trajectory-disjoint Training, Development, and Test splits. Across the six paired tasks, Action Delay, and Door Traversability, models are evaluated on novel trajectories under dynamics conditions or rules represented in training. In contrast, Speed assesses continuous parameter generalization, with the main results evaluating unseen speeds held out within the training range (Appendix D). Each paired task provides up to 32,768 training pairs, alongside 256 balanced counterfactual pairs in each evaluation split (Table 4). Full dataset statistics, dynamics parameters, and task-specific split checks are detailed in Appendix D.

3.4. Evaluation Protocol and Metrics

Evaluating latent in-context dynamics adaptation requires disentangling genuine context utilization from predictive shortcuts. Because models can minimize prediction errors shared across dynamics conditions or regress to average dynamics without conditioning on interaction history, prediction error alone does not establish genuine adaptation. Furthermore, as demonstrated in Appendix F, constructed predictions barely perturbed from the mean can still achieve high classification accuracy. Under strictly frozen weights, we evaluate two core aspects: Adaptation Accuracy (A_{future}), testing whether predictions distinguish diverging physical futures, and History Sensitivity (History %, G_H), testing whether matching history reduces prediction error relative to mismatched history. We complement these with Multi-Step Prediction Error (E/B), evaluating trajectory rollout fidelity against an unconditioned mean-future reference, and Normalized Response Error (NRE), measuring the accuracy of the model's conditional response

Table 1. Overview of the nine tasks evaluating in-context dynamics adaptation across four environments. History length h counts context observations (including the current observation) provided to the model. Sampling details and simulation settings appear in Appendices A and D.

Task	Environment	History length h	What history helps the model predict
Speed	TwoRoom	3	How far the agent moves under the same action
Pusher displacement	PushT	3	Whether the same action moves the pusher a shorter or longer distance
Arm mass	Reacher	3	How the same torque changes arm motion
Action delay	TwoRoom	7	When an action takes effect
Contact friction	PushT	3	How friction changes object motion during contact
Motion damping	PushT	3	How quickly object motion decays after contact ends
Gripper carry rule	Cube	3	Whether a closed gripper carries the cube
Door traversability	TwoRoom	3	Whether a visually identical door permits passage
Portal exit	TwoRoom	3	Where the agent emerges after entering the same portal

after removing condition-invariant errors.

Adaptation Accuracy (A_{future}). To test whether predictions distinguish diverging physical outcomes, we evaluate paired counterfactual predictions against latent ground-truth targets. For pair $i \in \{1, \dots, N\}$ with targets z_i^A, z_i^B and predictions $\hat{z}_i^A, \hat{z}_i^B$, a prediction is scored as correct only if it is strictly closer in MSE (d) to its corresponding future than to the alternative: $I_i^A = \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^A, z_i^B)]$, with I_i^B defined symmetrically:

$$A_{\text{future}} = \frac{100}{2N} \sum_{i=1}^{N} \left(I_i^A + I_i^B \right). \tag{2}$$

Under identical current observations and future actions, a deterministic model that ignores history produces identical predictions ($\hat{z}_i^A = \hat{z}_i^B$) and scores at most 50%, with ties counted as failures. Multi-condition tasks use task-specific scoring (Appendix E).

History Sensitivity (History %, G_H). To isolate whether error reductions stem specifically from context, we swap histories across dynamics conditions while holding the current observation, future actions, and target fixed. We report **History (%)** (the win rate where matched history yields strictly lower error than mismatched context) and normalized error reduction $G_H = (E_- - E_+)/B$, where E_- and E_+ are aggregate errors under mismatched and matched histories, respectively, and B is the aggregate error of the mean-future reference defined below.

Multi-Step Prediction Error (E/B). High one-step classification accuracy does not guarantee low trajectory rollout error, as minute perturbations toward the correct target can yield high accuracy (Appendix F). We therefore evaluate open-loop autoregressive rollouts across up to 25 simulator steps without intermediate ground-truth observations. Because checkpoints lack a shared physical metric, we normalize rollout error against an unconditioned mean-future reference [25]:

$$E/B = \frac{\sum_q L_q}{\sum_q B_q}. \tag{3}$$

For scene q, L_q denotes the mean squared rollout error averaged over K_q dynamics conditions, C_q shared action sequences, and $T = 5$ rollout steps; B_q is the corresponding error of the mean-future reference $\bar{z}_{qct} = K_q^{-1} \sum_k z_{qkct}$. For equally weighted conditions sharing the current observation and future actions, this reference minimizes squared error among predictions that do not distinguish conditions [18]; consequently, any deterministic model that ignores history cannot achieve $E/B < 1$. Changes in target representations must also be considered when comparing checkpoints.

Normalized Response Error (NRE). Prediction loss can decrease as errors shared across conditions shrink, even when the predicted difference between dynamics conditions remains inaccurate. To isolate the conditional response to differing physics, we define Normalized Response Error over true difference vectors $\Delta z_i = z_i^B - z_i^A$ and predicted differences $\Delta \hat{z}_i = \hat{z}_i^B - \hat{z}_i^A$:

$$\text{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}. \tag{4}$$

Subtracting predictions cancels condition-invariant errors. Predicting zero differential response yields $\text{NRE} = 1$, whereas $\text{NRE} = 0$ denotes an exact differential response.

Protocol Boundaries and Statistical Rigor. Latent metrics (E/B, NRE) are evaluated in each checkpoint's native

representation. We report means and standard deviations across training runs; trajectory-sampling uncertainty for the evaluated checkpoints is estimated via source-cluster bootstrap resampling, which does not reflect variability across training runs (Appendix E). Finally, to assess whether adaptation to varied dynamics impairs base control capabilities, downstream performance in unperturbed environments is evaluated via CEM planning (Section 4).

4. Experiments

4.1. Experimental Setup

Evaluated models and representation strategies. In-context dynamics adaptation relies on a predictor operating over latent representations. To investigate how different latent world model formulations exhibit this ability, we evaluate three action-conditioned architectures spanning distinct representation-learning strategies: (1) DINO-WM [37], which trains its dynamics predictor atop frozen, general-purpose DINOv2 features; (2) PLDM [30], which jointly optimizes visual representations and latent dynamics; and (3) LeWM [24], which combines latent prediction with explicit representation regularization to prevent representation collapse. During evaluation, all model weights remain strictly frozen. Because these models differ across architectures, loss formulations, and pretraining distributions, their comparison provides an empirical survey across established world-modeling choices rather than an ablation of a single structural factor.

Training schemes and design rationale. To examine how training choices affect adaptation, we evaluate four training regimes (Table 2):

Pretraining and data composition: Models trained exclusively under fixed dynamics (Original) establish whether adaptation emerges without training on ContextWorld data. To support adaptation while maintaining original-environment performance, we construct Mixed datasets by sampling trajectory windows equally $(1:1)$ from the original datasets and ContextWorld. Comparing models trained directly on this mixture (Mixed) against fine-tuned checkpoints (Original $\rightarrow$ Mixed) evaluates the benefit of base-dynamics pretraining, though this two-stage regime incorporates an additional training phase and differs in total updates.

Visual encoder updates: Full joint fine-tuning allows visual representations to adapt alongside the dynamics predictor, but may also alter latent feature geometry as dynamics vary. For LeWM and PLDM, we branch from the same pretrained checkpoint into Joint (updating all modules) and Frozen Encoder (fixing the visual backbone while keeping the projector, action encoder, and predictor trainable). This comparison tests whether freezing the visual encoder improves history-conditioned prediction, without holding the entire target representation static.

Models are trained independently for each task. To assess whether training on multi-dynamics distributions compromises base control capabilities, we measure task performance in unperturbed environments using Cross-Entropy Method (CEM) planning [12]. Optimization details and hyperparameters are provided in Appendix G.

Table 2. Training schemes. Mixed samples trajectory windows equally $(1:1)$ from original-environment and ContextWorld datasets to support adaptation while maintaining base performance. Comparing Mixed with Original $\rightarrow$ Mixed evaluates the benefit of pretraining under base dynamics, while Joint versus Frozen Encoder evaluates the effect of freezing the visual backbone. DINO-WM retains its frozen pretrained visual encoder across all schemes.

Scheme	Start	Data	Encoder
Original	Initial	Original	Updated†
Mixed	Initial	Mixed	Updated†
Original $\rightarrow$ Mixed (Joint)	Original	Mixed	Updated
Original $\rightarrow$ Mixed (Frozen Encoder)	Original	Mixed	Frozen

Initial weights are random for LeWM and PLDM. DINO-WM uses a pretrained visual encoder and a random predictor. †DINO-WM keeps its encoder frozen in every scheme; its continued-training scheme is labeled Original $\rightarrow$ Mixed. Other modules remain trainable; native training losses are retained.

4.2. Adaptation Results

Limited in-context dynamics adaptation under fixed dynamics. We first evaluate whether latent world models trained exclusively in fixed-dynamics environments adapt to varied physics without parameter updates. Across the seven binary tasks, models evaluated under the Original scheme score near or below the 50% history-agnostic upper bound (Table 3). Because a deterministic model that ignores history receives identical current observations and planned actions in paired counterfactuals, it cannot exceed 50%; however, these scores indicate that models trained under fixed dynamics show limited ability to distinguish futures across differing dynamics.

Mixed-data training improves adaptation but exposes task-dependent trade-offs. Introducing ContextWorld data via Mixed training improves adaptation on several tasks, but its effectiveness varies across models and tasks. Furthermore, no single architecture excels across all evaluated tasks. For instance, LeWM and PLDM achieve near-perfect performance on action delay ($> 97\%$), whereas DINO-WM reaches only 16.65%. Conversely, on pusher displacement, DINO-WM leads at 97.66%, while LeWM and PLDM remain much closer to the 50% reference. All

Table 3. **Adaptation metrics on the Test split.** Each cell lists Accuracy (%) ↑, History (%) ↑, and NRE ↓ from top to bottom. Entries are means; run counts and standard deviations appear in Table 6. Speed uses the same scoring rule for Accuracy and History. Bold marks the highest displayed Accuracy or History per task. NRE is measured in each checkpoint's target representation. A dash denotes no comparable checkpoint.

Model	Training scheme	Speed	Pusher displacement	Arm mass	Action delay	Contact friction	Motion damping	Gripper carry rule	Door traversability	Portal exit
LeWM	Original	58.04 58.04 0.896	48.96 28.32 1.018	50.07 51.63 0.977	—	49.93 50.13 1.001	50.00 48.63 1.007	52.21 73.76 0.909	50.33 50.94 0.996	50.07 51.95 0.990
	Mixed	98.19 98.19 0.042	50.78 79.10 0.974	85.74 89.84 0.045	97.70 99.18 0.013	50.00 49.41 1.000	50.00 48.24 1.004	50.00 49.80 1.000	99.83 **100.00** 0.041	50.78 76.37 0.922
	Original → Mixed (Joint)	96.89 96.89 0.055	91.80 99.02 0.215	87.30 93.36 0.038	—	50.39 54.30 0.999	50.00 49.61 1.001	50.00 52.34 0.997	99.83 **100.00** 0.038	50.39 66.80 0.934
	Original → Mixed (Frozen Encoder)	**98.44** **98.44** 0.030	97.46 98.83 0.128	77.15 86.52 0.346	—	**86.13** **92.77** 0.429	53.91 61.91 0.958	**76.37** **84.57** 0.271	**100.00** **100.00** 0.036	77.93 94.34 0.264
PLDM	Original	48.41 48.41 0.922	45.57 9.38 1.058	51.82 66.99 0.921	—	50.00 49.09 1.014	50.26 50.52 0.947	50.78 57.42 0.937	50.06 51.50 0.997	50.07 51.50 0.996
	Mixed	96.78 96.78 0.084	56.25 68.16 0.835	**91.02** **93.55** 0.031	**100.00** **100.00** 0.006	50.00 48.63 1.001	58.20 55.86 2.435	49.80 50.78 0.997	**100.00** **100.00** 0.022	72.27 79.10 0.128
	Original → Mixed (Joint)	98.33 98.33 0.047	59.18 73.44 0.740	84.77 90.82 0.073	—	50.00 46.48 1.020	**79.88** **81.84** 0.237	50.20 54.30 0.982	**100.00** **100.00** 0.023	52.54 61.52 0.708
	Original → Mixed (Frozen Encoder)	96.44 96.44 0.062	**98.63** **99.80** 0.101	81.84 89.26 0.213	—	73.63 83.59 0.595	54.49 64.26 0.867	52.93 57.81 0.606	**100.00** **100.00** 0.048	91.21 98.83 0.191
DINO-WM	Original	57.11 57.11 0.919	38.48 9.38 1.127	50.78 61.33 0.977	—	50.39 47.66 1.015	49.80 49.61 1.014	50.00 50.59 1.000	50.00 64.33 0.985	50.78 56.45 0.996
	Mixed	59.41 59.41 0.961	97.66 99.61 0.394	78.32 88.09 0.673	16.65 19.45 1.000	84.18 92.19 0.648	56.45 65.23 1.042	60.16 77.34 0.892	**100.00** **100.00** 0.217	**100.00** **100.00** 0.142
	Original → Mixed	84.78 84.78 0.411	—	—	—	—	55.27 61.13 1.111	62.89 80.27 0.871	—	—

three architectures attain near-perfect accuracy on door traversability. These rank reversals demonstrate that adaptation capabilities depend heavily on the specific dynamic variation, underscoring the need to benchmark world models across diverse physical settings rather than a single representative task.

History benefits do not necessarily imply an accurate conditional response. For LeWM on pusher displacement, Mixed training achieves a 79.10% History win rate, yet NRE remains 0.974, close to the zero-response reference (Table 3). Matching history can therefore reduce prediction error without accurately reproducing the difference between dynamics conditions.

Single-step history sensitivity does not guarantee low multi-step rollout error. A critical consideration is whether models that effectively utilize context in single-step settings maintain accurate rollouts over extended horizons. For the separate analysis on Development, LeWM and PLDM trained under Original → Mixed (Frozen Encoder) show strong one-step history sensitivity on pusher displacement, exceeding a 98% History win-rate and substantially reducing NRE (Appendix H). However, evaluating autoregressive rollouts across extended action sequences reveals a different picture: as shown in Figure 2, prediction errors divided by the error of the unconditioned mean-future reference (E_t/B_t) escalate monotonically, exceeding 1.0 from the third rollout step onward and resulting in aggregate E/B ratios above 1.4 for both models. These extended queries differ from the one-step tasks by lengthening both the planned action sequences and the rollout horizon. Consequently, while these checkpoints outperform the Original baseline, high one-step history sensitivity does not guarantee low multi-step prediction error under open-loop rollout.

4.3. Effects of Training Choices

Freezing the pretrained visual encoder helps some tasks and hurts others. For PLDM, it improves contact friction Test accuracy from 50.00% to 73.63% relative to joint training, but reduces motion damping accuracy from 79.88% to

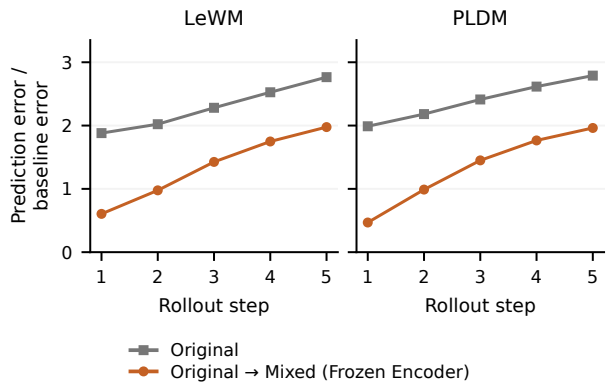


Figure 2. **Multi-step prediction on pusher displacement (Development).** Each rollout step spans five simulator steps simulated open-loop without intermediate observations. Prediction error is normalized by the error of predicting the mean future representation across dynamics conditions (E_t/B_t), computed within each checkpoint's native representation and averaged over available training runs. While checkpoints trained under Original → Mixed (Frozen Encoder) exhibit over 98% one-step history win-rates, their normalized multi-step rollout error steadily increases and exceeds the unconditioned mean-future reference ($E_t/B_t > 1.0$) from the third rollout step onward. These extended queries lengthen both the action sequences and prediction horizons relative to the one-step tasks; complete nine-task summaries appear in Appendix H.

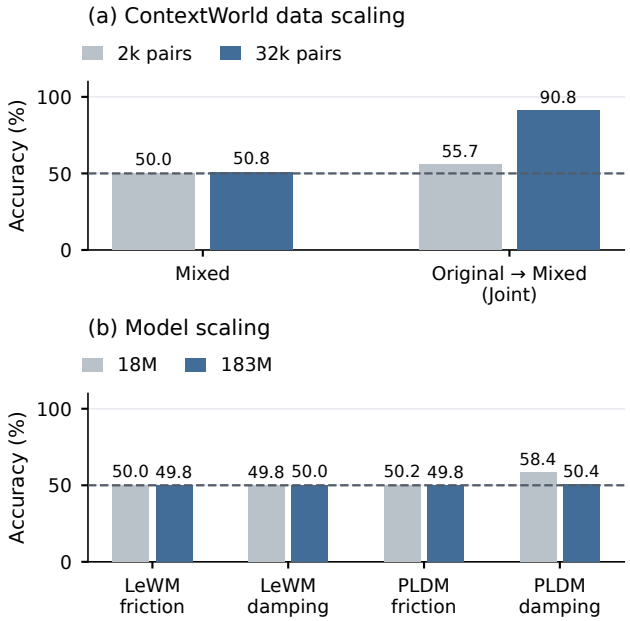


Figure 3. **Data and model scaling on Development.** (a) LeWM on pusher displacement, with 2,048 versus 32,768 ContextWorld training pairs and the same original-environment dataset. (b) LeWM and PLDM under Mixed. Labels above the bars give accuracy (%). Dashed lines mark the 50% paired-task reference. Epoch counts are fixed. Full results and run counts appear in Appendix H.1–H.2.

54.49% (Table 3). Each comparison holds the data, initialization, and training duration fixed. Because projectors and action encoders remain trainable, this intervention does not isolate a static target representation.

Figure 3 examines whether more data or capacity resolves weak adaptation. For LeWM on pusher displacement, data scaling substantially benefits Original → Mixed (Joint), whereas Mixed stays near chance. NRE also improves with joint fine-tuning (Appendix H.1). Fixed epochs allow update counts to change with data size, so this is not a fixed-compute comparison.

Under the 10-epoch recipe, expanding LeWM and PLDM from 18M to 183M parameters yields no consistent accuracy gain. PLDM on damping loses 8.01 percentage points of Development accuracy and 10.67 points of original-environment CEM success; CEM success decreases in all four comparisons (Appendix H.2). The cause of this regression remains unresolved.

4.4. Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10 (Figure 4).

On friction, PLDM joint training reduces prediction loss

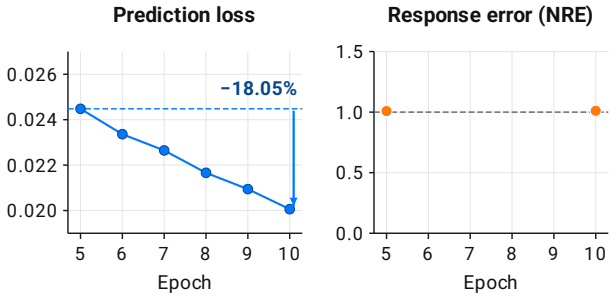


Figure 4. **Lower prediction loss does not ensure an accurate conditional response.** PLDM on contact friction, Original → Mixed (Joint). Recorded prediction loss falls by 18.05% from epochs 5 to 10, while training-query NRE stays near the zero-response reference of 1. Loss uses mixed training batches; NRE is measured on fixed paired training queries at the two endpoints. Full scheme and task comparisons appear in Figure 5.

by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high (Figure 5). Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping, with Development showing the same task-dependent ordering.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions. Optimizing the former does not ensure an accurate conditional response. This finding identifies insufficient constraints on history-conditioned dynamics in the evaluated training objectives and strategies.

5. Conclusion

ContextWorld establishes a standardized benchmark to evaluate in-context dynamics adaptation in action-conditioned latent visual world models under strictly frozen weights. Across nine tasks spanning four simulated environments, our paired history comparisons decouple whether models condition their predictions on interaction history from how accurately they forecast the resulting futures. Systematic evaluations of LeWM, PLDM, and DINO-WM demonstrate that while training on multi-dynamics data can improve adaptation, relative model advantages and the impact of visual encoder freezing remain task-dependent. Crucially, our training-distribution diagnostics expose a disconnect between prediction-loss optimization and in-context adaptation: conditional responses can remain poorly captured even as training loss steadily decreases or approaches convergence. These findings identify insufficient constraints on history-conditioned dynamics within the evaluated training objectives and strategies, underscoring the need for direct adaptation benchmarks and formulations that explicitly constrain history-conditioned responses.

Our findings are framed by specific diagnostic boundaries. Predictions are evaluated in each model's native representation rather than a shared physical error space; most tasks examine adaptation among dynamics conditions represented in training, while Speed evaluates unseen parameter values; and mixed-data checkpoints are trained independently per task, with planning experiments quantifying capability retention in unperturbed environments. These boundaries outline clear avenues for future research: establishing complementary cross-model evaluation in grounded physical coordinates, scaling toward generalist multi-task world models that generalize across broader dynamics, and examining closed-loop planning to test whether history-conditioned latent predictions translate directly into improved decision-making.

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A. Input Sampling

Observations are typically sampled every five simulator steps. The actions executed between consecutive observations are included in the interaction history.

B. Trajectory Construction and Pairing

Continuous trajectories. Each sample follows a continuous simulator trajectory from its history through its future, without intermediate resets or state overrides. Identical current observations may still conceal differences in velocity or contact state, so we perform task-specific checks of physical-state matching at the start of prediction. For action delay, we retain the actual queue of pending actions.

Starting positions. For tasks in which the two dynamics conditions cannot naturally share a starting position, we exchange their starting positions during generation and verify that the two conditions have identical sets of starting images. This check reduces the possibility that the starting image alone reveals the condition.

Predictions without history. Consider a complete, balanced pair with distinguishable encoded targets. A deterministic model that ignores history receives the same current observation and future action sequence for both samples and therefore produces the same prediction. This prediction cannot be strictly closer to each sample's own target than to the other target. Consequently, such a model scores at most 50% on the paired future-selection metric in Eq. 2; ties count as failures.

C. Action Delay Response Groups

The delay data retain all 11 delays from 0 to 10 simulator steps. At the five-step horizon used for the main one-step score, delays of 5–10 produce the same stationary response and are grouped together. This score therefore uses six distinguishable response groups: 0, 1, 2, 3, 4, and 5–10.

D. Dataset Statistics and Splits

D.1. Dataset Sizes

Table 4 lists the task data used in the training and evaluation comparisons. Development supports method and checkpoint selection; Test is reserved for final evaluation. Training sizes exclude the original-environment data used for mixed training. A pair contains two trajectories with different dynamics conditions; a trajectory count refers to recorded episodes, not to the number of training windows sampled from them. An evaluation case fixes the current observation and future action sequence and can include histories and targets from several dynamics conditions.

The action delay counts refer to the dataset covering all delays from 0 to 10 simulator steps. Its Training split contains 5,120 base cases, each simulated under all 11 delays, giving 56,320 trajectories. Development and Test each contain 300 base cases, again evaluated under all 11 delays. Door Training contains 7,680 trajectories for each of the passable and blocked rules.

Several examples can share a source episode: pusher displacement can use multiple action windows from one episode, and forward and mirrored damping examples can share a source. The 10,000 gripper carry training pairs use 10,000 distinct source episodes.

D.2. Paired Dynamics Conditions

Pusher displacement uses action gains of 60 and 140. Contact friction uses coefficients 0.05 and 0.80 between the pusher and the object. Motion damping fixes effective contact friction at 0.25 and uses simulator damping settings 0.2 and 1.0; here 0.2 gives faster decay, while 1.0 applies no additional decay. Arm mass uses densities of 500 and 1,500 for the MuJoCo arm link and finger geometry; the hand link retains its original density. Gripper carry changes whether the closed gripper carries the cube. Portal exit uses two exit rules for the same entrance, with paired trajectories reaching the shared query state before entering it. The numeric settings are simulator parameters, not additional model inputs.

D.3. Speed Values and Door Positions

Table 5 lists the speed values used for Training and evaluation. Development and Test use the same evaluation speeds but different interaction trajectories. Each split contains 300 cases for each of the 14 evaluated speeds, or 4,200 cases in total before comparing histories from different speeds. The main task results use the unseen speeds within the training range.

Door position is specified by an integer coordinate along the wall. Test uses the 42 positions $30 + 4k$, where $k = 0, \ldots, 41$. Development uses the 16 positions 65, 77, 79, 87, 112, 115, 131, 152, 167, 169, 171, 172, 175, 180, 183, 185. Training uses the integers from 36 through 188 after excluding all Development and Test positions and the additional unused positions 40, 127, and 132, leaving 96 positions. The three position sets are disjoint. Both passable and blocked rules are used across the splits; the test distinction is the door position, not a new rule.

D.4. Split Checks and Model Inputs

Split checks use source episode identifiers, trajectory template identifiers, or observation hashes, as appropriate for each task. Speed Development uses independently generated initial configurations and excludes initial states and current observations that duplicate Test cases. For Door, we

Table 4. Dataset sizes and condition coverage.

Task	Training	Development	Test
Speed	16,384 trajectories; 32 speeds	300 cases per speed	300 cases per speed
Pusher displacement	32,768 pairs	256 pairs	256 pairs
Arm mass	32,768 pairs	256 pairs	256 pairs
Action delay	56,320 trajectories; 11 delays	300 cases $\times$ 11 delays	300 cases $\times$ 11 delays
Contact friction	32,768 pairs	256 pairs	256 pairs
Motion damping	32,768 pairs	256 pairs	256 pairs
Gripper carry rule	10,000 pairs	256 pairs	256 pairs
Door traversability	15,360 trajectories; 96 door positions	300 cases; 16 door positions	300 cases; 42 door positions
Portal exit	32,768 pairs	256 pairs	256 pairs

Table 5. Speed values in the benchmark. The four evaluation groups are used in both Development and Test. Interpolation values lie within the Training range but are absent from Training.

Setting	Speed values
Training	2.6, 2.7, 2.8, 2.9, 3.1, 3.3, 3.5, 3.7, 3.8, 3.9, 4.1, 4.2, 4.4, 4.5, 4.7, 5.0, 5.1, 5.3, 5.5, 5.7, 5.9, 6.1, 6.2, 6.3, 6.6, 6.7, 6.8, 7.0, 7.2, 7.3, 7.8, 7.9
Seen speeds	3.1, 5.1, 7.0
Interpolation	3.4, 4.8, 6.9
Lower extrapolation	1.75, 1.95, 2.15, 2.35
Upper extrapolation	8.25, 8.75, 9.5, 10.25

also check the position sets listed above for overlap.

Hidden dynamics values, simulator states, and pair identifiers are retained for data generation, validation, and scoring. They are not passed to the prediction model, which receives only the visual observations and actions defined in Sec. 3.1.

E. Evaluation Details

Paired tasks. The strict comparisons in Eq. 2 do not count ties as correct. The two comparisons within a pair are dependent. Appendix B explains the upper bound for a deterministic model that ignores history.

Speed. For each target speed, accuracy is the proportion of cases in which matching history gives strictly lower prediction error than every alternative history. We then average accuracy equally across target speeds. Ties do not count as correct: if all histories produce the same prediction, accuracy is zero. The main results use the interpolation speeds listed in Appendix D.

Action delay. We select the candidate true future with the lowest mean squared error, resolving ties in favor of the smaller delay. The prediction is correct when the selected delay and the true delay belong to the same response group. Within each case, we first average correctness over delays in each group, then average equally across the six groups defined in Appendix C. The final accuracy is the mean over cases.

Door traversability. For a prediction made with matching history, we compare the mean squared errors to the passage and blockage futures. The prediction is correct only if the error to the true future is strictly smaller than the error to the alternative future. Accuracy is averaged over cases and the two rules; ties do not count as correct.

Multi-step prediction horizons. Each prediction step advances five simulator steps. We measure prediction errors at 5, 10, 15, 20, and 25 simulator steps after the current observation.

Within-scene aggregation. Let z_{qkct} and $\hat{z}_{qkct}$ denote the true and predicted future representations for scene q, dynamics condition k, action sequence c, and prediction time t. With K_q conditions, C_q shared action sequences, and $T = 5$ prediction times, the errors used in Eq. 3 are

$$L_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$

$$B_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2,$$

$$\bar{z}_{qct} = \frac{1}{K_q} \sum_{k} z_{qkct}.$$

L2World tests history dependence by shuffling context observations [33]; our controls instead swap histories between dynamics conditions. For history controls, E_+ and E_- sum the matching- and mismatched-history scene errors using the same averaging over conditions, action sequences, and

prediction times. When a sample has multiple mismatched histories, their errors are averaged equally. A positive G_H indicates that matching history reduces error; it does not imply a low complete prediction error.

Mean-future reference. The reference in Eq. 3 is used only for scoring and is never provided to the model. The ratio is defined only when $\sum_q B_q > 0$. When conditions share the current observation and future actions and are equally weighted, a deterministic model that ignores history cannot achieve $E/B < 1$: it must make the same prediction across conditions, and the mean-future reference minimizes the error of such a prediction.

Response error settings. We specify the prediction horizon for each NRE analysis. The ratio in Eq. 4 is defined only when $\sum_i \|\Delta z_i\|^2 > 0$. NRE equal to 1 does not by itself show that a model ignores history. Errors shared by both predictions cancel in the difference, so an exact conditional response does not imply exact complete predictions.

Uncertainty estimates. For multi-step errors, we use cluster bootstrap resampling [15, 17] at the level of source trajectories or scenes, retaining related samples together. These intervals describe scene-sampling uncertainty for the evaluated checkpoints, not variability across training runs.

Interpreting multiple metrics. The metrics describe different aspects of the same predictions and are not independent pieces of evidence.

F. Scoring Controls

We validate the paired scorer using constructed predictions on 256 Development pusher-displacement scenes, separately in the target representations of LeWM Original and Original → Mixed (Frozen Encoder). The queries use a nonzero action candidate and the five-step prediction horizon; the predictor is not run for this check. Both representations give the results below. With $m = (z_A + z_B)/2$, even predictions $m + 0.001(z_k - m)$ select the correct future while recovering only a tiny fraction of the true response. This control explains why we report response error alongside task accuracy. It verifies the scorer on these queries, not physical calibration across all representations.

Constructed prediction	Accuracy (%)	History (%)	NRE
True target for each condition	100	100	0
Same target z_A for both conditions	50	0	1
Same midpoint m for both conditions	0	0	1
Swapped true targets	0	0	4
$m + 0.001(z_k - m)$ for condition k	100	100	0.998001

G. Training and Diagnostic Settings

Optimization. The training recipe specifies 10 epochs per stage on eight GPUs, with 128 windows per device and no gradient accumulation, giving a global batch size of 1,024. LeWM and PLDM use AdamW with learning rate 5×10^{-5} and weight decay 0.001. DINO-WM uses learning rate 5×10^{-4} and zero weight decay; its pretrained DINOv2-small visual encoder remains frozen. The models retain their respective prediction and regularization losses. The Original → Mixed schemes load model weights but reset optimizer state. Encoder freezing fixes both parameters and training-state buffers; it does not freeze the projector or action encoder.

Input and sampling. The standard setup uses three observation frames, five simulator steps between frames, and a one-frame prediction target. Action delay mixed training instead uses seven-frame history. The original-environment data are the TwoRoom, PushT, Reacher, and Cube datasets released with LeWM; each task uses only the dataset for its environment. Only the ContextWorld Training split is mixed into training. For component lengths N_{orig} and N_{CW}, the mixture has a nominal epoch length of $2\max(N_{\mathrm{orig}}, N_{\mathrm{CW}})$ windows before partitioning and batching, with equal counts from the two components. The smaller component is sampled repeatedly. Consequently, equal epoch counts do not guarantee equal update counts across data scales, and the two-stage schemes include additional original-environment training.

Prediction diagnostics. The main task comparison uses Test. The multi-step and history diagnostics below use Development, with the same checkpoint identities where the input protocol is comparable. Multi-step evaluation covers 2,436 source scenes across the nine tasks and predicts at 5, 10, 15, 20, and 25 simulator steps using only the initial observed history. The candidate actions and multi-step targets extend the one-step task queries. Figure 2 reports E_t/B_t: prediction and mean-future errors are summed over queries separately at each horizon, normalized within each checkpoint, then averaged equally over training repetitions. The aggregate E/B instead combines errors over all horizons before division; it is not the arithmetic mean of these per-horizon ratios. Each checkpoint is scored separately, and matching training repetitions are then averaged equally. Intervals use 4,000 source-cluster bootstrap samples; all conditions, actions, and times from a source remain together. They condition on the evaluated checkpoints and do not estimate variation from retraining the model. Original action-delay checkpoints do not have the native seven-frame interface and are excluded from this comparison. DINO-WM has no jointly updated visual-encoder scheme; its comparable Original → Mixed results cover Speed, damping, and gripper carry.

Planning evaluation. CEM results measure success in the original environments, with no hidden-dynamics history intervention. Most checkpoint evaluations use six evaluation seeds with 50 episodes each; some original LeWM and PLDM references use three sets of 100 episodes or an aggregated 300-episode result. Evaluation repetitions are pooled within each checkpoint, then training repetitions are averaged equally. A single original-environment checkpoint supplies the reference for all tasks from that environment; these repeated entries are not independent experiments.

Training-query analysis. The friction and damping comparisons use 32,768 training pairs, identical original-environment initialization within each task, and the same 50/50 mixture, batch size, and duration. We evaluate 512 fixed training pairs

and 256 Development pairs per task at epochs 5 and 10, predicting the next observation five simulator steps ahead. Loss summaries use the logged mixed-training batches, approximately 50–70 records per epoch, rather than every training batch. Checkpoints and logs are aligned by update step. The curves show recorded means and batch interquartile ranges; loss medians give the same monotonic direction for friction joint training from epochs 5 to 10. NRE is evaluated in each checkpoint's target representation and is not a common physical error across checkpoints. For damping with the encoder frozen, prediction loss changes by only -0.39% between these endpoints and fluctuates in between. The late-training plateau is consistent with the loss approaching convergence.

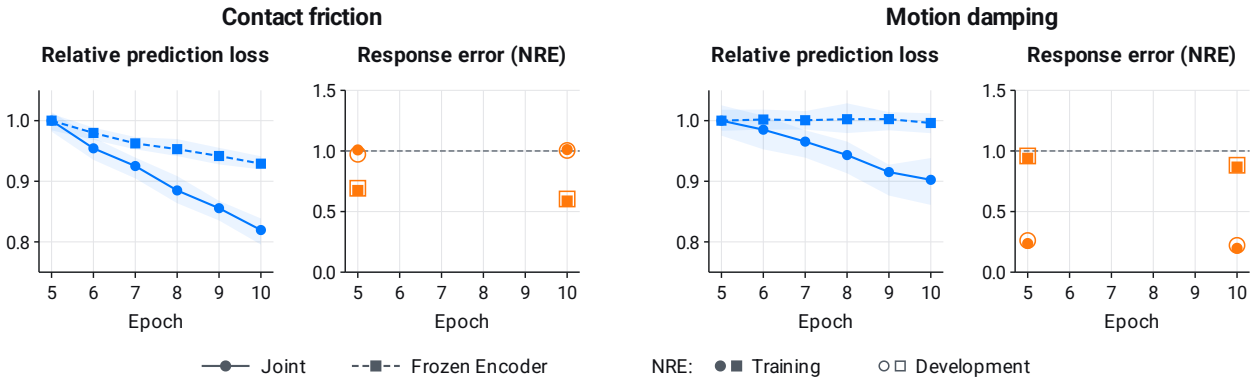


Figure 5. **Training loss and conditional responses across tasks and schemes.** Within each task, PLDM Joint and Frozen Encoder share Original initialization. Loss is normalized to epoch 5; shading shows batch interquartile ranges. NRE contains separate Training and Development measurements at epochs 5 and 10; 1 marks the zero-response reference. Loss uses mixed batches, whereas NRE uses fixed paired task queries.

H. Additional Results

Table 6 reports all three Test adaptation metrics with training-run counts and standard deviations. Table 7 reports the separate Development diagnostics, multi-step prediction errors, and original-environment planning results. History measures strict error reduction from matching history on the task queries; NRE measures the conditional response on those queries. E/B uses the extended multi-step action panels and includes errors shared by both conditions. Because the query sets and prediction horizons differ, these columns should not be treated as measurements of the same event. All latent errors are computed in the checkpoint's own target representation.

Table 6. **Complete Test adaptation metrics.** Each row uses the same evaluated checkpoints for all three metrics. Entries are means $\pm$ sample standard deviations across n training runs; a single run has no sample standard deviation. Accuracy and History are percentages; NRE is dimensionless. Speed uses unseen interpolation speeds, and its Accuracy and History use the same scoring rule. Action delay uses seven-frame history. Only available comparable configurations are listed.

Task	Model	Training scheme	n	Accuracy ↑	History ↑	NRE ↓
Speed	LeWM	Original	3	58.04 ± 1.70	58.04 ± 1.70	0.896 ± 0.027
Speed	LeWM	Mixed	3	98.19 ± 0.57	98.19 ± 0.57	0.042 ± 0.011
Speed	LeWM	Original → Mixed (Joint)	1	96.89	96.89	0.055
Speed	LeWM	Original → Mixed (Frozen Encoder)	1	98.44	98.44	0.030
Speed	PLDM	Original	3	48.41 ± 4.33	48.41 ± 4.33	0.922 ± 0.026
Speed	PLDM	Mixed	3	96.78 ± 2.02	96.78 ± 2.02	0.084 ± 0.013
Speed	PLDM	Original → Mixed (Joint)	1	98.33	98.33	0.047
Speed	PLDM	Original → Mixed (Frozen Encoder)	1	96.44	96.44	0.062
Speed	DINO-WM	Original	1	57.11	57.11	0.919
Speed	DINO-WM	Mixed	3	59.41 ± 3.87	59.41 ± 3.87	0.961 ± 0.011

Task	Model	Training scheme	n	Accuracy ↑	History ↑	NRE ↓
Speed	DINO-WM	Original → Mixed	1	84.78	84.78	0.411
Pusher displacement	LeWM	Original	3	48.96 ± 1.00	28.32 ± 12.76	1.018 ± 0.016
Pusher displacement	LeWM	Mixed	1	50.78	79.10	0.974
Pusher displacement	LeWM	Original → Mixed (Joint)	1	91.80	99.02	0.215
Pusher displacement	LeWM	Original → Mixed (Frozen Encoder)	1	97.46	98.83	0.128
Pusher displacement	PLDM	Original	3	45.57 ± 1.88	9.38 ± 4.20	1.058 ± 0.012
Pusher displacement	PLDM	Mixed	1	56.25	68.16	0.835
Pusher displacement	PLDM	Original → Mixed (Joint)	1	59.18	73.44	0.740
Pusher displacement	PLDM	Original → Mixed (Frozen Encoder)	1	98.63	99.80	0.101
Pusher displacement	DINO-WM	Original	1	38.48	9.38	1.127
Pusher displacement	DINO-WM	Mixed	1	97.66	99.61	0.394
Arm mass	LeWM	Original	3	50.07 ± 0.11	51.63 ± 0.60	0.977 ± 0.002
Arm mass	LeWM	Mixed	1	85.74	89.84	0.045
Arm mass	LeWM	Original → Mixed (Joint)	1	87.30	93.36	0.038
Arm mass	LeWM	Original → Mixed (Frozen Encoder)	1	77.15	86.52	0.346
Arm mass	PLDM	Original	3	51.82 ± 1.30	66.99 ± 9.68	0.921 ± 0.016
Arm mass	PLDM	Mixed	1	91.02	93.55	0.031
Arm mass	PLDM	Original → Mixed (Joint)	1	84.77	90.82	0.073
Arm mass	PLDM	Original → Mixed (Frozen Encoder)	1	81.84	89.26	0.213
Arm mass	DINO-WM	Original	1	50.78	61.33	0.977
Arm mass	DINO-WM	Mixed	1	78.32	88.09	0.673
Action delay	LeWM	Mixed	3	97.70 ± 0.59	99.18 ± 0.38	0.013 ± 0.002
Action delay	PLDM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.006 ± 0.001
Action delay	DINO-WM	Mixed	3	16.65 ± 0.03	19.45 ± 0.11	1.000 ± 0.000
Contact friction	LeWM	Original	3	49.93 ± 0.23	50.13 ± 0.63	1.001 ± 0.004
Contact friction	LeWM	Mixed	1	50.00	49.41	1.000
Contact friction	LeWM	Original → Mixed (Joint)	1	50.39	54.30	0.999
Contact friction	LeWM	Original → Mixed (Frozen Encoder)	1	86.13	92.77	0.429
Contact friction	PLDM	Original	3	50.00 ± 0.34	49.09 ± 1.58	1.014 ± 0.010
Contact friction	PLDM	Mixed	1	50.00	48.63	1.001
Contact friction	PLDM	Original → Mixed (Joint)	1	50.00	46.48	1.020
Contact friction	PLDM	Original → Mixed (Frozen Encoder)	1	73.63	83.59	0.595
Contact friction	DINO-WM	Original	1	50.39	47.66	1.015
Contact friction	DINO-WM	Mixed	1	84.18	92.19	0.648
Motion damping	LeWM	Original	3	50.00 ± 0.00	48.63 ± 1.60	1.007 ± 0.009
Motion damping	LeWM	Mixed	1	50.00	48.24	1.004
Motion damping	LeWM	Original → Mixed (Joint)	1	50.00	49.61	1.001

Task	Model	Training scheme	n	Accuracy ↑	History ↑	NRE ↓
Motion damping	LeWM	Original → Mixed (Frozen Encoder)	1	53.91	61.91	0.958
Motion damping	PLDM	Original	3	50.26 ± 0.30	50.52 ± 0.63	0.947 ± 0.007
Motion damping	PLDM	Mixed	1	58.20	55.86	2.435
Motion damping	PLDM	Original → Mixed (Joint)	1	79.88	81.84	0.237
Motion damping	PLDM	Original → Mixed (Frozen Encoder)	1	54.49	64.26	0.867
Motion damping	DINO-WM	Original	1	49.80	49.61	1.014
Motion damping	DINO-WM	Mixed	1	56.45	65.23	1.042
Motion damping	DINO-WM	Original → Mixed	1	55.27	61.13	1.111
Gripper carry rule	LeWM	Original	3	52.21 ± 1.11	73.76 ± 1.97	0.909 ± 0.029
Gripper carry rule	LeWM	Mixed	1	50.00	49.80	1.000
Gripper carry rule	LeWM	Original → Mixed (Joint)	1	50.00	52.34	0.997
Gripper carry rule	LeWM	Original → Mixed (Frozen Encoder)	1	76.37	84.57	0.271
Gripper carry rule	PLDM	Original	3	50.78 ± 0.20	57.42 ± 1.35	0.937 ± 0.007
Gripper carry rule	PLDM	Mixed	1	49.80	50.78	0.997
Gripper carry rule	PLDM	Original → Mixed (Joint)	1	50.20	54.30	0.982
Gripper carry rule	PLDM	Original → Mixed (Frozen Encoder)	1	52.93	57.81	0.606
Gripper carry rule	DINO-WM	Original	1	50.00	50.59	1.000
Gripper carry rule	DINO-WM	Mixed	1	60.16	77.34	0.892
Gripper carry rule	DINO-WM	Original → Mixed	1	62.89	80.27	0.871
Door traversability	LeWM	Original	3	50.33 ± 0.00	50.94 ± 2.84	0.996 ± 0.003
Door traversability	LeWM	Mixed	3	99.83 ± 0.29	100.00 ± 0.00	0.041 ± 0.018
Door traversability	LeWM	Original → Mixed (Joint)	1	99.83	100.00	0.038
Door traversability	LeWM	Original → Mixed (Frozen Encoder)	1	100.00	100.00	0.036
Door traversability	PLDM	Original	3	50.06 ± 0.10	51.50 ± 0.67	0.997 ± 0.002
Door traversability	PLDM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.022 ± 0.005
Door traversability	PLDM	Original → Mixed (Joint)	1	100.00	100.00	0.023
Door traversability	PLDM	Original → Mixed (Frozen Encoder)	1	100.00	100.00	0.048
Door traversability	DINO-WM	Original	1	50.00	64.33	0.985
Door traversability	DINO-WM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.217 ± 0.003
Portal exit	LeWM	Original	3	50.07 ± 0.11	51.95 ± 0.70	0.990 ± 0.003
Portal exit	LeWM	Mixed	1	50.78	76.37	0.922
Portal exit	LeWM	Original → Mixed (Joint)	1	50.39	66.80	0.934
Portal exit	LeWM	Original → Mixed (Frozen Encoder)	1	77.93	94.34	0.264
Portal exit	PLDM	Original	3	50.07 ± 0.11	51.50 ± 0.30	0.996 ± 0.001
Portal exit	PLDM	Mixed	1	72.27	79.10	0.128
Portal exit	PLDM	Original → Mixed (Joint)	1	52.54	61.52	0.708
Portal exit	PLDM	Original → Mixed (Frozen Encoder)	1	91.21	98.83	0.191
Portal exit	DINO-WM	Original	1	50.78	56.45	0.996
Portal exit	DINO-WM	Mixed	1	100.00	100.00	0.142

Table 7. Development prediction diagnostics and original-environment CEM. Entries average the available training repetitions; E/B intervals are 95% source-bootstrap intervals. History and CEM are percentages. Scheme names match the main text.

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Speed	LeWM	Original	59.15	0.896	1.498	[1.478, 1.517]	92.44
Speed	LeWM	Mixed	98.00	0.042	0.363	[0.345, 0.381]	97.44
Speed	LeWM	Original → Mixed (Joint)	96.67	0.055	0.406	[0.384, 0.429]	97.00
Speed	LeWM	Original → Mixed (Frozen Encoder)	98.56	0.030	0.144	[0.132, 0.156]	95.00
Speed	PLDM	Original	48.48	0.925	3.377	[3.309, 3.445]	90.56
Speed	PLDM	Mixed	97.19	0.084	0.520	[0.498, 0.544]	96.89
Speed	PLDM	Original → Mixed (Joint)	98.89	0.046	0.353	[0.327, 0.379]	96.33
Speed	PLDM	Original → Mixed (Frozen Encoder)	95.56	0.065	0.302	[0.285, 0.319]	98.00
Speed	DINO-WM	Original	58.56	0.928	1.973	[1.954, 1.991]	100.00
Speed	DINO-WM	Mixed	56.56	0.967	2.101	[2.070, 2.133]	99.89
Speed	DINO-WM	Original → Mixed	86.00	0.411	0.886	[0.847, 0.927]	100.00
Pusher displacement	LeWM	Original	34.05	1.015	2.378	[2.284, 2.481]	82.22
Pusher displacement	LeWM	Mixed	73.63	0.981	3.572	[3.421, 3.741]	78.67
Pusher displacement	LeWM	Original → Mixed (Joint)	97.46	0.222	2.504	[2.425, 2.586]	86.00
Pusher displacement	LeWM	Original → Mixed (Frozen Encoder)	98.05	0.144	1.464	[1.396, 1.537]	73.67
Pusher displacement	PLDM	Original	15.89	1.048	2.471	[2.395, 2.552]	75.67
Pusher displacement	PLDM	Mixed	66.60	0.896	6.533	[6.241, 6.847]	61.33
Pusher displacement	PLDM	Original → Mixed (Joint)	73.83	0.798	3.811	[3.664, 3.968]	72.00
Pusher displacement	PLDM	Original → Mixed (Frozen Encoder)	99.61	0.115	1.445	[1.388, 1.506]	59.33
Pusher displacement	DINO-WM	Original	14.65	1.113	3.300	[3.166, 3.446]	76.67
Pusher displacement	DINO-WM	Mixed	98.63	0.432	1.996	[1.941, 2.055]	76.67
Arm mass	LeWM	Original	50.98	0.977	687.381	[571.339, 816.332]	55.89
Arm mass	LeWM	Mixed	91.60	0.048	51.291	[45.684, 56.751]	58.00
Arm mass	LeWM	Original → Mixed (Joint)	91.80	0.039	47.152	[41.802, 52.374]	59.00
Arm mass	LeWM	Original → Mixed (Frozen Encoder)	88.28	0.338	31.814	[27.581, 35.547]	59.67
Arm mass	PLDM	Original	66.80	0.922	18.594	[16.630, 20.929]	69.67
Arm mass	PLDM	Mixed	92.19	0.033	13.617	[12.780, 14.631]	51.67
Arm mass	PLDM	Original → Mixed (Joint)	91.02	0.077	44.421	[38.659, 50.208]	58.67
Arm mass	PLDM	Original → Mixed (Frozen Encoder)	92.38	0.211	16.644	[15.067, 18.031]	89.00
Arm mass	DINO-WM	Original	59.38	0.980	48.246	[45.632, 51.198]	90.67
Arm mass	DINO-WM	Mixed	89.26	0.658	14.246	[13.770, 14.655]	75.00
Action delay	LeWM	Mixed	99.00	0.012	1.462	[1.439, 1.487]	87.33

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Action delay	PLDM	Mixed	99.99	0.007	1.807	[1.795, 1.820]	90.89
Action delay	DINO-WM	Mixed	21.14	1.000	65.764	[64.835, 66.676]	33.56
Contact friction	LeWM	Original	50.59	1.001	2.753	[2.680, 2.834]	82.22
Contact friction	LeWM	Mixed	50.00	0.999	2.693	[2.642, 2.747]	76.67
Contact friction	LeWM	Original → Mixed (Joint)	57.23	0.999	2.714	[2.674, 2.758]	86.00
Contact friction	LeWM	Original → Mixed (Frozen Encoder)	92.77	0.448	2.891	[2.833, 2.953]	83.33
Contact friction	PLDM	Original	49.28	1.012	3.869	[3.753, 3.992]	75.67
Contact friction	PLDM	Mixed	50.00	1.001	6.269	[5.975, 6.602]	65.67
Contact friction	PLDM	Original → Mixed (Joint)	51.37	1.004	7.086	[6.828, 7.366]	73.33
Contact friction	PLDM	Original → Mixed (Frozen Encoder)	85.94	0.604	4.499	[4.391, 4.618]	79.67
Contact friction	DINO-WM	Original	48.63	1.019	6.109	[5.850, 6.372]	76.67
Contact friction	DINO-WM	Mixed	93.16	0.684	3.673	[3.603, 3.744]	75.67
Motion damping	LeWM	Original	49.09	0.995	5.308	[5.042, 5.616]	82.22
Motion damping	LeWM	Mixed	49.61	1.002	291.614	[253.071, 335.959]	78.00
Motion damping	LeWM	Original → Mixed (Joint)	50.20	1.001	95.787	[83.792, 110.333]	88.33
Motion damping	LeWM	Original → Mixed (Frozen Encoder)	58.79	0.997	8.202	[7.657, 8.832]	84.33
Motion damping	PLDM	Original	51.69	0.957	5.973	[5.668, 6.318]	75.67
Motion damping	PLDM	Mixed	58.59	1.521	25.424	[22.201, 29.444]	61.00
Motion damping	PLDM	Original → Mixed (Joint)	79.10	0.221	8.247	[7.957, 8.547]	70.00
Motion damping	PLDM	Original → Mixed (Frozen Encoder)	62.11	0.882	5.553	[5.276, 5.871]	78.33
Motion damping	DINO-WM	Original	52.34	1.001	8.465	[8.122, 8.825]	76.67
Motion damping	DINO-WM	Mixed	66.21	1.021	7.273	[7.077, 7.482]	76.67
Motion damping	DINO-WM	Original → Mixed	65.62	1.083	6.927	[6.731, 7.135]	76.00
Gripper carry rule	LeWM	Original	74.80	0.911	11.678	[10.728, 12.783]	65.44
Gripper carry rule	LeWM	Mixed	49.61	1.001	198.472	[166.622, 241.013]	61.33
Gripper carry rule	LeWM	Original → Mixed (Joint)	52.34	0.998	13.085	[11.613, 14.946]	71.00

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Gripper carry rule	LeWM	Original → Mixed (Frozen Encoder)	78.12	0.320	11.937	[11.096, 12.883]	67.00
Gripper carry rule	PLDM	Original	57.62	0.923	15.350	[12.126, 22.152]	53.44
Gripper carry rule	PLDM	Mixed	52.54	0.985	147.591	[121.468, 184.796]	55.00
Gripper carry rule	PLDM	Original → Mixed (Joint)	54.49	0.985	26.553	[24.128, 29.592]	56.67
Gripper carry rule	PLDM	Original → Mixed (Frozen Encoder)	58.01	0.594	29.969	[21.538, 46.388]	56.67
Gripper carry rule	DINO-WM	Original	49.80	1.001	42.857	[41.696, 44.068]	72.67
Gripper carry rule	DINO-WM	Mixed	80.66	0.890	11.789	[11.454, 12.141]	71.67
Gripper carry rule	DINO-WM	Original → Mixed	82.03	0.874	11.747	[11.417, 12.082]	73.67
Door traversability	LeWM	Original	45.72	0.997	4.676	[4.568, 4.784]	92.44
Door traversability	LeWM	Mixed	100.00	0.045	2.981	[2.938, 3.024]	95.11
Door traversability	LeWM	Original → Mixed (Joint)	100.00	0.047	2.427	[2.378, 2.478]	97.67
Door traversability	LeWM	Original → Mixed (Frozen Encoder)	100.00	0.036	2.442	[2.400, 2.483]	94.00
Door traversability	PLDM	Original	52.22	0.997	7.520	[7.323, 7.718]	90.56
Door traversability	PLDM	Mixed	100.00	0.021	2.404	[2.384, 2.424]	95.78
Door traversability	PLDM	Original → Mixed (Joint)	100.00	0.027	2.349	[2.302, 2.397]	99.33
Door traversability	PLDM	Original → Mixed (Frozen Encoder)	100.00	0.049	2.455	[2.419, 2.491]	98.67
Door traversability	DINO-WM	Original	61.00	0.986	4.233	[4.143, 4.321]	100.00
Door traversability	DINO-WM	Mixed	100.00	0.223	1.329	[1.303, 1.354]	97.56
Portal exit	LeWM	Original	51.82	0.990	822.254	[587.576, 1303.713]	92.44
Portal exit	LeWM	Mixed	77.54	0.918	58.552	[47.450, 74.465]	98.33
Portal exit	LeWM	Original → Mixed (Joint)	66.21	0.919	61.538	[49.099, 81.373]	95.67
Portal exit	LeWM	Original → Mixed (Frozen Encoder)	93.16	0.253	13.286	[10.625, 17.147]	89.33
Portal exit	PLDM	Original	51.63	0.996	1056.976	[789.611, 1513.078]	90.56
Portal exit	PLDM	Mixed	78.71	0.121	37.556	[35.991, 39.195]	97.00
Portal exit	PLDM	Original → Mixed (Joint)	62.11	0.694	21.881	[20.065, 24.077]	98.67
Portal exit	PLDM	Original → Mixed (Frozen Encoder)	98.24	0.203	10.485	[9.219, 12.101]	98.67
Portal exit	DINO-WM	Original	54.69	0.995	41.991	[40.909, 43.144]	100.00
Portal exit	DINO-WM	Mixed	100.00	0.143	8.092	[8.005, 8.181]	100.00

H.1. Data Scale

These comparisons use Development and hold training epochs fixed. The sample counts describe pairs, not necessarily independent source episodes. Pusher displacement uses 2,048, 8,192, and 32,768 pairs; friction and damping compare 8,192 with 32,768; mass and portal exit compare 2,048 with 32,768. Gripper carry expands from 2,048 to 10,000 source episodes. Portal expansion also broadens initial distances and action magnitudes, so its comparison changes coverage as well as size.

Table 8. Data-scale comparisons on Development. n counts training repetitions contributing to task accuracy and NRE. Planning repetitions can differ and are not counted as additional training runs.

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Pusher displacement	LeWM	Mixed	2,048	3	50.00	1.000
Pusher displacement	LeWM	Mixed	32,768	1	50.78	0.981
Pusher displacement	LeWM	Original → Mixed (Joint)	2,048	1	55.66	0.810
Pusher displacement	LeWM	Original → Mixed (Joint)	8,192	1	82.42	0.390
Pusher displacement	LeWM	Original → Mixed (Joint)	32,768	1	90.82	0.222
Pusher displacement	PLDM	Mixed	2,048	3	50.07	1.003
Pusher displacement	PLDM	Mixed	32,768	1	56.64	0.896
Pusher displacement	DINO-WM	Mixed	2,048	3	89.19	0.583
Pusher displacement	DINO-WM	Mixed	32,768	1	96.09	0.432
Arm mass	LeWM	Mixed	2,048	3	85.35	0.132
Arm mass	LeWM	Mixed	32,768	1	87.11	0.048
Arm mass	PLDM	Mixed	2,048	3	52.60	1.054
Arm mass	PLDM	Mixed	32,768	1	91.21	0.033
Arm mass	DINO-WM	Mixed	2,048	3	51.37	0.999
Arm mass	DINO-WM	Mixed	32,768	1	80.47	0.658
Contact friction	LeWM	Mixed	8,192	3	50.07	1.000
Contact friction	LeWM	Mixed	32,768	1	50.00	0.999
Contact friction	PLDM	Mixed	8,192	3	50.13	1.001
Contact friction	PLDM	Mixed	32,768	1	50.20	1.001
Contact friction	DINO-WM	Mixed	8,192	3	49.61	1.025
Contact friction	DINO-WM	Mixed	32,768	1	81.84	0.684
Motion damping	LeWM	Mixed	8,192	3	50.07	0.999
Motion damping	LeWM	Mixed	32,768	1	49.80	1.002
Motion damping	PLDM	Mixed	8,192	3	51.89	2.122
Motion damping	PLDM	Mixed	32,768	1	58.40	1.521
Motion damping	DINO-WM	Mixed	8,192	3	50.13	1.014
Motion damping	DINO-WM	Mixed	32,768	1	57.42	1.021
Gripper carry rule	LeWM	Mixed	2,048	3	49.93	1.001
Gripper carry rule	LeWM	Mixed	10,000	1	50.00	1.001

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Gripper carry rule	PLDM	Mixed	2,048	3	50.00	0.996
Gripper carry rule	PLDM	Mixed	10,000	1	50.00	0.985
Gripper carry rule	DINO-WM	Mixed	2,048	3	50.33	0.995
Gripper carry rule	DINO-WM	Mixed	10,000	1	61.33	0.890
Portal exit	LeWM	Mixed	2,048	3	50.91	0.925
Portal exit	LeWM	Mixed	32,768	1	51.95	0.918
Portal exit	PLDM	Mixed	2,048	3	69.53	0.155
Portal exit	PLDM	Mixed	32,768	1	73.63	0.121
Portal exit	DINO-WM	Mixed	2,048	3	99.41	0.554
Portal exit	DINO-WM	Mixed	32,768	1	100.00	0.143

H.2. Model Capacity

Both model sizes use the same 32,768-pair datasets, training seed, Mixed initialization, and 10-epoch recipe. The larger model changes the visual backbone from ViT-Tiny to ViT-Base, the representation width from 192 to 768, and predictor depth from six to nine layers. This is a joint expansion of the encoder and predictor, not a predictor-only intervention.

		Accuracy (%)		NRE		CEM (%)	
Task	Model	18M	183M	18M	183M	18M	183M
Contact friction	LeWM	50.00	49.80	0.999	1.000	76.67	74.00
Motion damping	LeWM	49.80	50.00	1.002	1.000	78.00	71.33
Contact friction	PLDM	50.20	49.80	1.001	1.107	65.67	47.33
Motion damping	PLDM	58.40	50.39	1.521	1.536	61.00	50.33

The four endpoint comparisons do not show a consistent improvement from increasing capacity under this recipe. Only measured checkpoints are used in the training-curve analysis: epochs 1–10 for the large LeWM models and epochs 5 and 10 for the other models. There is no shared trajectory of accurate training responses followed by worsening Development response error across these runs. This does not exclude earlier overfitting where checkpoints are unavailable, or gains from retuning the larger models.